Assignment 1: TCP/IP methods and Attack Methods

Deadline: Friday 11/9 17:00

This assignment can be done by at most two persons.

Name1: _____

Name2: _____

A. TCP/IP protocol and Security

This section contains questions related to the security issues that emerge from the TCP/IP protocol.

1. Why is the IP protocol unreliable?

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2. IP is unreliable, and TCP uses IP. How does TCP provide reliable service to the application layer?

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3. What does TCP do if the message to be sent is larger than what a single datagram can handle?

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4. What are the minimum and maximum header size of IP packets?

   …

5. An IP packet arrives at a router with the first eight bits as 01000011. The router discards the packet. Why?

   …

6. Why is it necessary to have both IP address and port number in a packet?

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7. Which of the protocols TCP, UDP and IP provides for reliable communication?

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B. Scanning Attacks

A scanning attack is a common type of attack based on the TCP/IP protocol. The following questions aim at understanding how these attacks can be done.

8. What is the purpose of host scanning?

…

9. How does ping scanning work?

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10. Why are ping scans often not effective?

…

11. Why are SYN/ACK scans done?

…

12. How may hosts respond to SYN/FIN messages?

…

13. How does Traceroute (or Tracert) work?

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14. Why is port scanning done?

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15. How does TCP port scanning work?

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16. Why is sending a long stream of scanning messages dangerous for attackers?

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17 How do attackers use stealth scanning to reduce danger in the previous question?

…

18. What rules would you add to the firewall to prevent the SYN/ACK attack?

…

19. How many packets would be sent by an attacker to port scan 100 hosts for all well-known ports?

…

C. Attack Methods Based on TCP/IP Protocol

Besides scanning attacks there is a large variety of attacks based on the TCP/IP protocol. This section aims at understanding some of the most popular, the technique used and the consequences of the attack.

20. What is fingerprinting?

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21. Distinguish between active and passive fingerprinting.

…

22. Describe SYN flooding attack.

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23. Which measures can be deployed to avoid a SYN flooding attack?

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24. Describe how SYN cookies can be used to stop a SYN flooding attack.

…

25. Describe the Smurf attack.

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26. Describe DDoS attacks.

…

27. List some of the attacks that do use IP address spoofing.

…

28. List some of the attacks that do <u>not</u> use IP address spoofing.

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